

Examiner reports

Q1.

This was quite a long question, combining a test of elementary probability with a standard request for the mean and variance of a random variable. Most candidates scored very highly on the question as a whole. A common mistake occurred in part (b)(ii), where many candidates confused 'neither' with 'not both' and subtracted their from $\frac{1}{9}$. Another error, made by only a small number of candidates, was to divide the correct value of the mean by 5.

Q2.

This proved more demanding than probability questions on recent papers. Although most candidates made some progress, relatively few included all the possibilities in part (a)(iii). There was also confusion between the number of copies sold and the number of copies in stock. For example in part (b)(ii) if one copy was sold by Sherrats the probability of two being sold by Hughes is not 0.18, as used by many candidates, but $0.18 + 0.24$.

Q3.

Nearly all candidates identified the correct binomial distribution and, although a few confused 'more than 3' with '3 or more', the majority gained full marks.

Q4.

Candidates found this question more challenging than they have found probability questions in the past. While part (a) presented no problems, in part (b) candidates found it difficult to combine the data in the table with the fact that Veronica had reviewed 10 comedies, 15 dramas and 5 other films.

Q5.

Many candidates scored full marks in parts (a) and (c) but could not answer part (b), where $1 - \text{answer to part (a)}$ was required. The conditional probability in part (d) caused the usual problems for many candidates although a significant number of candidates scored full marks on this question.

Q6.

Parts (a), (b) and (c) were answered well but many candidates failed to include 0.5 in their calculations for part (d).

Q7.

Several perfect solutions were seen from candidates who understood the laws of probability but most candidates struggled with this probability question and often missed the point that the events involved were not independent. A significant number of candidates did not attempt the question.

Q8.

This was a relatively high scoring question, with the most common error being to omit 2, 3 and 6 from parts (a)(ii), (b)(ii) and (b)(iii) respectively.

Q9.

Many candidates scored full marks in parts (a)(i) and (ii) but several found parts (iii) and (iv) hard and assumed independence of events when this was not true. Many found the conditional probabilities in parts (v) and (vi) difficult but there were a considerable number of fully correct solutions. A large number of candidates answered part (b) correctly but some were inaccurate in their description of the event in simple terms. Some stated that the event $(R \cap H \cap T)$ referred to a male who is aged 55-74 or who has high blood pressure or who is being treated, despite obtaining the correct answer for $P(R \cap H \cap T)$.

Q10.

Although most candidates were able to answer parts (a) and (b), few were able to deal with the given conditions in the other parts. Many, for example, in part (c) calculated the probability of Louise winning the first game and winning the series by three games to one, instead of taking Louise's winning the first game as given.

Q11.

This challenging probability question was exceptionally well answered by a significant number of candidates. There were many candidates scoring full marks.

The proof in part (a) was sometimes done by candidates' assuming both 0.28 and 0.2 to be true rather than by showing that if $P(B) = 0.1$ then $P(C) = 0.2$. Also some candidates showed that if $P(C) = 0.2$ then $P(B) = 0.1$, which was allowed.

Part (b) was not always correctly answered but most candidates showed a good understanding of conditional probability.

Q12.

Most made a good start but many, although having the right idea in part (b)(ii), made errors in the detail.

Q13.

Most candidates were able to identify the binomial distribution and successfully use the tables. However, only the stronger candidates were able to answer part (c) by correctly combining the probabilities calculated earlier in the question.

Q14.

Candidates had few problems with this question.

Q15.

The intention of part (a) was to provide a straightforward lead into the question, but it

proved to be the least well answered part. A common answer was $1/22$. Otherwise the question was well answered. A few candidates answered 'with replacement' and others failed to include 3 in part (d) or 6 in part (e).

Q16.

A common mistake on part (a)(iii) was to multiply the probability of choosing a junior member by the probability of choosing a female member. Part (b) proved testing. A good proportion were able to determine whether or not events were mutually exclusive. Those who attempted to explain why V and W were not independent usually gave an explanation in rather vague, general terms, rather than the required numerical demonstration.

Q17.

Many candidates scored full marks in parts (a), (b) and (c) but most found the conditional probability in part (d) difficult to find. A surprising number failed to get the correct answer to parts (a) and (b) but answered part (c) correctly!

Q18.

Most candidates scored good marks on part (a). In part (b), the most common error was to carry out the calculation with replacement.

Q19.

Most candidates could start this question but relatively few completed it. Part (c) proved particularly testing.

Q20.

Many candidates scored full marks and most made a very good effort. There were still a few candidates who clearly avoided questions on probability.

Q21.

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Q22.

Most candidates scored highly on this question; they were evidently comfortable tackling probability questions.

Q23.

Most candidates coped with part (a). The most common errors in parts (b) and (c) were to use 'with replacement' and/or to ignore the different ways of the outcome occurring. Part (d) was poorly answered by the majority of candidates.

Q24.

There were many arithmetic errors in this question, some caused by slips in converting percentages into probabilities. Although the question asked for probabilities, many candidates felt obliged to give the answers as percentages - presumably because



percentages were used in the stem. This is not incorrect but did lead to yet more arithmetic slips.

Many candidates failed to answer the question asked in part (b), instead answering similar questions of their own devising. Although part (b) was intended as a step on the way to answering part (c), it was common for incorrect answers to part (b) to be followed by a correct answer to part (c).



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